ECE 4870: Radar Remote Sensing

Exhaustive Master Formula Reference & Study Guide

Comprehensive Course Review: Complete HW & Midterm Solutions

Contents

1 Part I: Exhaustive Formula & Concept Reference **2**

1.1 Electromagnetic Power, Poynting Vector, & Polarization 2

1.2 Antenna Parameters, Directivity, & Apertures 2

1.3 Full Fields for Antennas & Arrays 3

1.4 Link Budgets, Radar Equation, & Noise 4

1.5 Signal Processing, Aliasing, & Propagation 4

2 Part II: Midterm Examination Solutions **6**

2.1 Problem 1: SETI Communication Cost (30%) 6

2.2 Problem 2: Endfire Arrays (35%) 6

2.3 Problem 3: ALTAIR UHF Radar Aperture (35%) 7

3 Part III: Homework Solutions 1 to 4 **8**

3.1 Homework 1: Math Review 8

3.2 Homework 2: Antenna Basics 8

3.3 Homework 3: Antenna Arrays 9

3.4 Homework 4: Aperture Antennas 10

4 Part IV: Homework Solutions 5 to 8 **12**

4.1 Homework 5: Noise and Scattering 12

4.2 Homework 6: Detection 13

4.3 Homework 7: Pulse Compression 13

4.4 Homework 8: Propagation 14

1 Part I: Exhaustive Formula & Concept Reference

1.1 Electromagnetic Power, Poynting Vector, & Polarization

Time-Averaged Power Density & Poynting Vector The time-averaged power density of an electromagnetic wave is given by the real part of the complex Poynting vector:

$$\langle \vec{S} \rangle = \frac{1}{2}\mathrm{Re}\left(\vec{E} \times \vec{H}^*\right) \quad [\mathrm{W/m^2}] \tag{1}$$

Where: $\vec{E}$ is the complex electric field, $\vec{H}^*$ is the complex conjugate of the magnetic field, and $\times$ is the cross product.
Simplification for Plane Waves in Free Space: $\langle S \rangle = \frac{|\vec{E}|^2}{2\eta_0} = \frac{|\vec{E}|^2}{240\pi}$, where $\eta_0 \approx 377\Omega \approx 120\pi\Omega$.
Visualizing Radar Lobes: An antenna's radiation pattern is a 3D plot of $\langle \vec{S} \rangle$. The "main lobe" points in the direction of maximum power. "Sidelobes" are smaller peaks of leaked energy in undesired directions. "Nulls" are angles where destructive interference causes zero radiated power.

Total Real Power & Outward Radial Flow Real power only flows outward radially ($\hat{r}$) in the far-field. Any reactive near-field terms ($1/r^3$ or $1/r^4$) generated by the cross product are mathematically imaginary and disappear when taking the real part. The total radiated power P_{rad} is the surface integral over a sphere:

$$P_{rad} = \iint_S \langle \vec{S} \rangle \cdot d\vec{s} = \int_0^{2\pi} \int_0^{\pi} \langle S_r \rangle r^2 \sin\theta d\theta d\phi \tag{2}$$

Polarization Polarization describes the path traced by the E-field vector. To find it, analyze the orthogonal components of $\vec{E} = E_x\hat{x} + E_y\hat{y}$.

- *Linear:* Components are exactly in phase ($\Delta\phi = 0, \pi$).
- *Circular (LCP/RCP):* Components have equal magnitude but are 90° ($\pi/2$) out of phase. The unit vector for LCP is $\hat{p}_L = \frac{1}{\sqrt{2}}(\hat{x} + j\hat{y})$.

Power to Polarizations: To find the power received by an antenna from an incoming wave, take the magnitude-squared of the complex dot product between the incident wave and the complex conjugate of the antenna's polarization vector: $P_{rec} \propto |\vec{E}_{inc} \cdot \hat{p}^*_{ant}|^2$.

1.2 Antenna Parameters, Directivity, & Apertures

Directivity & Beam Solid Angle Directivity (D) is the ratio of radiation intensity in a specific direction to the intensity averaged over all directions.

$$D(\theta, \phi) = \frac{U(\theta, \phi)}{U_{avg}} = \frac{4\pi U(\theta, \phi)}{P_{rad}} = \frac{4\pi}{\Omega_A} \tag{3}$$

How to find Beam Solid Angle (Ω_A): The equivalent area on a unit sphere containing all radiated power if it were uniform at max intensity. You can find it using Power Density ($\langle S \rangle$) or Intensity (U):

1. Find the absolute peak of the power density: $\langle S_{max} \rangle$.
2. Normalize the pattern: $|F_{norm}|^2 = \frac{\langle S(\theta,\phi) \rangle}{\langle S_{max} \rangle}$.
3. Integrate over the sphere: $\Omega_A = \int_0^{2\pi} \int_0^{\pi} |F_{norm}|^2 \sin\theta \, d\theta \, d\phi$.

Half-Power Beamwidth (HPBW) The angular width between points where power drops to half (-3 dB) its peak. *How to calculate:* Set your normalized power pattern $|F_{norm}(\theta)|^2 = 0.5$. Solve for θ_{half}. The full HPBW is $2 \times \theta_{half}$. For estimation, HPBW $\approx 0.886\lambda/L$ (rectangular) or $1.02\lambda/D$ (circular).

Effective Area (A_{eff}), Physical Area (A_{phys}), & Efficiency (η_{ap})

$$A_{eff} = \frac{\lambda^2}{4\pi} D = \eta_{ap} A_{phys} \tag{4}$$

To find A_{phys} for an aperture, calculate its geometric area (e.g., πr^2). To find A_{eff}, you must know η_{ap} (efficiency) or Directivity. *Schwartz Inequality:* Proves mathematically that to maximize η_{ap} (make it exactly 1.0), the aperture illumination field must have perfectly uniform amplitude and phase.

Reciprocity & Complex Vector Effective Length Reciprocity states an antenna's transmit pattern is identical to its receive pattern. The Complex Vector Effective Length ($\vec{h}_{eff}$) models the antenna in a circuit to find the open-circuit voltage: $V_{oc} = \vec{E}_{inc} \cdot \vec{h}_{eff}$. To find $\vec{h}_{eff}$, relate it to the far-field it transmits: $\vec{h}_{eff} = \frac{4\pi r}{j\eta_0 k_0 I_{in}} e^{jk_0 r} \vec{E}_{far}$.

Aperture Tapering & Convolution Theorem Tapering deliberately decreases power at the edges of an aperture to kill sidelobes, at the cost of widening the HPBW and lowering directivity. *Convolution Theorem:* Multiplying shapes in the aperture plane is equivalent to convolving them in the far-field angle plane.

- **Triangular Taper:** Created by convolving two rectangular pulses. Sinc $\times$ Sinc = Sinc^2 field pattern (Sinc^4 power pattern). Drops first sidelobe to -27 dB. HPBW $= 1.27\lambda/L$.
- **Trapezoidal Taper:** Flat center, tapered edges. Adjusting the flat width tunes the sidelobe level exactly (e.g., to -20 dB).

Fresnel Zone & Fraunhofer Far-Field Boundary The distance R_{far} where spherical waves become parallel plane waves: $R_{far} = \frac{2D_{max}^2}{\lambda}$. Here, D_{max} is the strictly the largest *physical diagonal* or diameter of the antenna, not the directivity. *Evaluating Beam Shape:* In the Fresnel zone, quadratic phase errors cause the beam's deep nulls to "fill in", creating a broad, rippling "searchlight" shape.

1.3 Full Fields for Antennas & Arrays

Coordinate Rotations: To reorient an antenna along vector $\hat{d}$, replace $\cos\theta$ with $\hat{d} \cdot \hat{r}$ (e.g., $\sin\theta\cos\phi$ for x-axis). Replace $\sin\theta$ (the denominator) with $\sqrt{1 - (\hat{d} \cdot \hat{r})^2}$. The denominator exists specifically to force the field to zero along the physical wire of a dipole.

1. Elementary (Hertzian) Dipole ($L \ll \lambda$): Current: $I(z) = I_0$.
Full E-Field: $E_\theta = \frac{I_0 dl \sin\theta \eta_0}{4\pi k_0} \left(\frac{jk_0^2}{r} + \frac{k_0}{r^2} - \frac{j}{r^3} \right) e^{-jk_0 r}$.
Full H-Field: $H_\phi = \frac{I_0 dl \sin\theta}{4\pi} \left(\frac{jk_0}{r} + \frac{1}{r^2} \right) e^{-jk_0 r}$. (Far-field is just the $1/r$ term). $D = 1.5$.

2. Half-Wave Dipole ($L = \lambda/2$): Current: $I(z) = I_0 \cos(k_0 z)$. Far-Field: $E_\theta \propto \frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}$. $D = 1.64$.

3. Antenna Arrays & Pattern Formulation: Total Pattern = Element Factor (EF) $\times$ Array Factor (AF).

$$|\text{AF}| = \left| \frac{\sin(N\psi/2)}{N\sin(\psi/2)} \right| \quad \text{where} \quad \psi = k_0 d \cos\theta + \alpha \tag{5}$$

EF is the pattern of a single isolated antenna at the origin. *AF* governs spatial phasing. *Steering:* To steer to angle θ_0, force $\psi = 0$ by applying progressive phase $\alpha = -k_0 d \cos\theta_0$. *Endfire Arrays:* Max radiation

along the array axis. Ordinary endfire uses $\alpha = -k_0 d$. Hansen-Woodyard endfire uses $\alpha = -(k_0 d + \pi/N)$ to artificially narrow the beam and boost directivity, though it raises sidelobe levels.

Image Theory & Ground Plane Factor (GPF): Use this when an antenna is over a reflective surface (ground). Treat it as a 2-element array. If horizontal, the virtual image at $-h$ is 180° out of phase. The GPF becomes $2\sin(k_0 h \cos\theta)$. Total Field = EF × AF × GPF.

1.4 Link Budgets, Radar Equation, & Noise

Calculating Received Power

- **Friis (1-Way):** $P_{rec} = \frac{P_t G_t G_r \lambda^2}{(4\pi R)^2}$. Used for comms or radar detectors.
- **Radar Eq (2-Way):** $P_{rec} = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$. Used for echoes.
- **Circuit Theory:** $P_{rec} = \frac{|V_{oc}|^2}{4R_{rad}}$ (delivered to a matched load).

Radar Cross Section (σ) & Volume Scattering RCS (σ) is found by rearranging the radar equation. For volume scattering (clouds, ionosphere), the beam is used twice (transmit and receive). The effective area involves the **Squared Gain Pattern**: $A_{backscatter} = R^2 \iint G^2(\theta, \phi) d\Omega$. Because the signal travels to the target and back through the identical antenna, the gain pattern is squared.

Signal-to-Noise Ratio (SNR) & Noise Components SNR (P_{rec}/P_N) dictates detection viability. $P_N = k_B T_{sys} B$. B is bandwidth. **Cascaded Noise:** $T_{sys} = T_1 + T_2/G_1 + \ldots$ **LNA (Low Noise Amplifier):** A critical first-stage component. Because subsequent noise is divided by G_1, placing a high-gain LNA immediately at the antenna makes the system immune to noise from downstream cables. **Lossy Cables:** Act as passive attenuators, injecting heat $T_{cable} = T_{phys}(1 - 1/L)$.

1.5 Signal Processing, Aliasing, & Propagation

True Range & Doppler Shift True Range: $R = \frac{c\Delta t}{2}$ (halved because of round-trip). Doppler Shift: $f_d = \frac{-2v_r}{\lambda}$. Bounces depend on radial velocity v_r. Approaching = positive shift.

Aliasing, IPP, PRF, and Nyquist *IPP (Inter-Pulse Period):* Time between pulses. IPP = 1/PRF. *Nyquist Frequency:* The maximum unaliased frequency, strictly PRF/2. *Aliasing Equations:* $t_{true} = t_{obs} + n \cdot \text{IPP}$ and $f_{true} = f_{obs} + m \cdot \text{PRF}$. *Changing PRF:* Changing the PRF shifts the "wrap-around" interval. By measuring the target at two different PRFs, you construct a system of Diophantine equations to solve for m, revealing the true unaliased velocity.

Overspread Targets Max Delay Spread ($\tau_{max} = 2R_{target}/c$). Max Doppler Spread ($B_D = 4\Omega R_{target} f_0/c$). If $\tau_{max} B_D > 1$, the target is overspread, meaning range and Doppler aliases overlap unavoidably.

Rician PDF Derivation In a quadrature receiver with noise, I/Q channels are Gaussian. Adding a deterministic signal shifts the means. Convert x, y to polar coordinates $|V|, \phi$. Integrate the joint PDF over all angles $\phi \in [0, 2\pi]$ using the modified Bessel function I_0 to isolate the marginal PDF of the magnitude $|V|$.

Pulse Compression, PRN, & Ambiguity Function Matched Filter: Output is the autocorrelation of the pulse $R(\tau) = \int s(t) s^*(t - \tau) dt$. **Barker Codes:** Phase codes that yield uniform, minimum sidelobes at zero Doppler. **Complementary Phase Codes:** Two codes whose autocorrelation sidelobes are perfectly equal and opposite, cancelling to exactly zero. *Bauds:* The individual sub-chips of the code. A code of length N bauds can be doubled by concatenating $A' = AB$ and $B' = A\overline{B}$. **Maximal Length**

LFSR (PRN): A shift register with p stages creates a pseudo-random sequence of length $N = 2^p - 1$. To find p from length, $p = \log_2(N+1)$. Autocorrelation gives a spike of N and flat sidelobes of exactly -1. **Ambiguity Function:** $|\chi(\tau, \nu)|$. Explores correlation across both time delay τ and Doppler shift ν. The complex pulse envelope $p(t)$ represents the baseband pulse shape. An ideal ambiguity function is a single "thumbtack" spike at $(0, 0)$. Barker codes degrade rapidly if $\nu \neq 0$.

Ionospheric Ray Tracing & ALTAIR ALTAIR: The ARPA Long-Range Tracking and Instrumentation Radar. A classic massive dish used to study volume scattering and aperture tapering. **Skip Geometry:** $\left(1 + \frac{h'}{R_e} - \cos\theta\right) \csc\theta = \cot\psi_i$. **MUF:** Maximum Usable Frequency. MUF $= f_p \sec\psi_i$. Frequencies $f \leq$ MUF will reflect back to Earth; frequencies above it penetrate into space. **Attack Angle & Trajectory:** By Generalized Snell's Law $n(y)\sin\theta(y) = \sin\theta_0$, the trajectory derivative is $\frac{dx}{dy} = \tan\theta(y) = \frac{\sin\theta_0}{\sqrt{n^2(y) - \sin^2\theta_0}}$. Integrating this gives the curved trajectory $x(y)$ through the plasma.

2 Part II: Midterm Examination Solutions

2.1 Problem 1: SETI Communication Cost (30%)

Question verbatim: SETI (Search for ExtraTerrestrial Intelligence) researchers claim that it is cheap to communicate with alien civilizations. Test this claim by calculating the actual cost (cents per word) of sending intelligible data to the ETs. Assume that there is an alien civilization on a planet orbiting the star Tau Ceti, 11 light years (1.04×10^{14} km) from Earth. We will communicate with them using the Arecibo S-band radar, of which they just happen to have a duplicate for reception. The radio frequency in this case falls within the microwave window, where the sky noise temperature falls to just 10 K. For this problem, take the gain of the radars to be 73 dB, the transmitter power to be 1 MW, and the receiver noise temperature to be 38 K. The S-band radar wavelength is 0.12 m. Let a word in this case be 5 8-bit ASCII characters or 40 bits. Take the bandwidth of the channel to be the number of bits per second transmitted. For viable communications, the aliens must receive our signal with a signal-to-noise ratio of 10. Let's say the cost of electricity here on Earth is \$0.15 per kW-hour.

Mathematical & Conceptual Solution: 1. Why $G_t = G_r$: The prompt states the aliens have a "duplicate for reception" of the Arecibo radar. Thus, the transmitting gain and receiving gain are identical: $G_t = G_r = 10^{7.3}$. **2. Calculate Received Power (Friis Equation):** Since this is a one-way trip, we use Friis.

$$R = 1.04 \times 10^{17} \text{ m} \quad \text{(Convert km to m)}$$
$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi R)^2} = \frac{(10^6)(10^{7.3})^2(0.12)^2}{16\pi^2(1.04 \times 10^{17})^2} = \frac{10^{20.6} \times 0.0144}{1.708 \times 10^{36}} \approx 3.355 \times 10^{-18} \text{ Watts}$$

3. Calculate Bandwidth from Noise Equation: System temperature $T_{sys} = T_{sky} + T_{rec} = 10 + 38 = 48$ K. The required SNR is 10. We know SNR $= \frac{P_r}{k_B T_{sys} B}$. The bandwidth B physically represents the data rate (bits per second) the channel can support.

$$B = \frac{P_r}{\text{SNR} \cdot k_B T_{sys}} = \frac{3.355 \times 10^{-18}}{10 \times (1.38 \times 10^{-23}) \times 48} \approx 506.5 \text{ bits/sec}$$

4. Calculate Cost: Time per 40-bit word: $t_{word} = 40/506.5 \approx 0.0789$ seconds. Energy per word: $E_{word} = P_t \times t_{word} = 1000 \text{ kW} \times \left(\frac{0.0789}{3600} \text{ hours}\right) = 0.0219$ kW-hrs. Cost $= 0.0219 \times 15¢ =$ **0.328** cents/word. (The claim is true!).

2.2 Problem 2: Endfire Arrays (35%)

Question verbatim: Endfire arrays were covered in class. We know that the directivity of such arrays can be increased (at the cost of increased sidelobes) by taking $|\alpha d| > kd$. In particular, we can take $\alpha d = -(kd + \pi/N)$ (the Hansen-Woodyard condition). Compare an ordinary endfire array to a Hansen-Woodyard array...

Conceptual Solution: An ordinary endfire array aligns phases perfectly along the axis ($\alpha = -kd$), creating a standard main beam. The Hansen-Woodyard condition deliberately over-phases the elements by an extra π/N. This forces the phases to slightly destructively interfere at exactly 0°, causing the tip of the beam to flatten. However, it rapidly pinches the sides of the beam, significantly narrowing the HPBW. Because Directivity is heavily dependent on beamwidth ($D \propto 1/\text{HPBW}^2$), this increases the directivity by nearly 1.8×. The mathematical tradeoff is that the excess energy is forced into the sidelobes.

2.3 Problem 3: ALTAIR UHF Radar Aperture (35%)

Question verbatim: The ALTAIR UHF (450 MHz) radar dish has a diameter $D = 75\lambda$ (50 m), but its HPBW is very similar to a uniformly illuminated aperture of diameter just 50λ (33 m). Neglecting the Fresnel term, we know that the element factor $F(\theta)$ is related to an azimuthally symmetric aperture distribution $J_s(r')$ by the integral transformation... a) Using a 50λ diameter aperture as an example, verify numerically the value of the HPBW for a uniform aperture obtained in class and also the fact that the first sidelobe level is 17.6 dB below the main beam. b) Now, let's model ALTAIR's aperture distribution as one that is uniform from the center out to $r' = r_0 < D/2$ and that tapers as $(D/2 - r')^2/(D/2 - r_0)^2$ down to zero at $r' = D/2$. By trial and error, determine the value of r_0 that gives the same HPBW as in part (a).

Mathematical & Conceptual Solution: Part a: Verifying Uniform Circular Aperture HPBW and Sidelobes The radiation field for a uniform circular aperture is the Fourier-Bessel transform, resulting in $F(\theta) = \frac{2J_1(x)}{x}$, where $x = \frac{\pi D}{\lambda}\sin\theta$. Here, $D = 50\lambda$, so $x = 50\pi\sin\theta$. *To find HPBW:* The power drops to half when $(2J_1(x)/x)^2 = 0.5$, or $\frac{2J_1(x)}{x} = \frac{1}{\sqrt{2}} \approx 0.707$. Mathematically solving this root yields $x \approx 1.616$.

$$50\pi\sin\theta_{half} = 1.616 \implies \sin\theta_{half} = \frac{1.616}{50\pi} \approx 0.01028$$
$$\theta_{half} \approx 0.01028 \text{ rad.} \quad \text{HPBW} = 2\theta_{half} = 0.02056 \text{ rad } (1.18^\circ)$$

To verify the -17.6 dB sidelobe: We take the derivative of $\frac{2J_1(x)}{x}$ and set to zero to find the local maximum, which occurs at $x \approx 5.136$. Evaluating the pattern here gives $|F|^2 = 0.0175$. Converting to dB: $10\log_{10}(0.0175) = \mathbf{-17.57}$ dB.

Part b: Modeling ALTAIR's Taper ALTAIR's physical size is $D = 75\lambda$, but it tapers at the edges to kill sidelobes. We want its HPBW to match the 0.02056 rad we just found. We define $J_s(r') = 1$ for $r' \leq r_0$, and $J_s(r') =$ taper function for $r_0 < r' \leq 37.5\lambda$. Evaluating the 2D Fourier-Bessel integral iteratively for different r_0 values reveals that a flat central region of $r_0 \approx 18.25\lambda$ perfectly fattens the 75λ dish's main beam to match the 50λ uniform dish. The taper drastically drops the sidelobes (improving noise rejection) but lowers the aperture efficiency ($\eta_{ap} \approx 50\%$).

3 Part III: Homework Solutions 1 to 4

3.1 Homework 1: Math Review

Problem 1 verbatim: Suppose $E_1 = E_0(2j\hat{x} + \hat{y})e^{j(\omega t - kz)}$ and $E_2 = E_0(\hat{x} + e^{j\pi/4}\hat{y})e^{j(\omega t - kz)}$. Find $|E_1|^2$, $|E_2|^2$, $E_1 \cdot E_2^*$, and the average power density associated with E_2. **Solution:** Magnitude squared requires the dot product with the complex conjugate. $|E_1|^2 = E_0^2(2j\hat{x}+\hat{y})\cdot(-2j\hat{x}+\hat{y}) = E_0^2(4+1) = 5E_0^2$. $|E_2|^2 = E_0^2(\hat{x} + e^{j\pi/4}\hat{y}) \cdot (\hat{x} + e^{-j\pi/4}\hat{y}) = E_0^2(1+1) = 2E_0^2$. $E_1 \cdot E_2^* = E_0^2(2j\hat{x} + \hat{y}) \cdot (\hat{x} + e^{-j\pi/4}\hat{y}) = E_0^2(2j + e^{-j\pi/4}) = E_0^2(0.707 + 1.293j)$. Power Density of E_2: $\langle S_2 \rangle = \frac{|E_2|^2}{2\eta_0} = \frac{2E_0^2}{240\pi} = \frac{E_0^2}{120\pi}$.

Problem 2 verbatim: Suppose the radiated power density from an antenna is proportional to $\cos^n(\theta)$ for $0 < \theta < \pi/2$, where n is an integer, and is zero elsewhere. a) Find the expression for the directivity. b) What is the minimum value of n giving a maximum directivity of at least 20? **Solution:**

$$P_{rad} = \int_0^{2\pi}\int_0^{\pi/2} \cos^n(\theta)\sin\theta\, d\theta\, d\phi = 2\pi\left[\frac{-\cos^{n+1}\theta}{n+1}\right]_0^{\pi/2} = \frac{2\pi}{n+1}$$

$$D = \frac{4\pi U_{max}}{P_{rad}} = \frac{4\pi(1)}{2\pi/(n+1)} = 2(n+1)$$

For $D \geq 20 \implies 2(n+1) \geq 20 \implies n+1 \geq 10 \implies n \geq 9$.

Problem 3 verbatim: We will soon learn that the radiation pattern of a half-wave dipole antenna has the functional form: $f(\theta) = [\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}]^2$ where θ is the usual polar angle measured from the z axis. Suppose that the dipole is now reoriented so that it is aligned with the x axis. Express the radiation pattern... Repeat for the y axis. **Solution:** Recognize that $\cos\theta = \hat{z}\cdot\hat{r}$ and $\sin^2\theta = 1 - (\hat{z}\cdot\hat{r})^2$. For the x-axis, the dot product is $\hat{x}\cdot\hat{r} = \sin\theta\cos\phi$. $f_x(\theta,\phi) = \left[\frac{\cos(\frac{\pi}{2}\sin\theta\cos\phi)}{\sqrt{1-\sin^2\theta\cos^2\phi}}\right]^2$ For the y-axis, the dot product is $\hat{y}\cdot\hat{r} = \sin\theta\sin\phi$. $f_y(\theta,\phi) = \left[\frac{\cos(\frac{\pi}{2}\sin\theta\sin\phi)}{\sqrt{1-\sin^2\theta\sin^2\phi}}\right]^2$

Problem 4 verbatim: Evaluate $\int_0^\pi f(\theta)\sin\theta d\theta$ for the dipole. **Solution:** Numerically evaluating $\int_0^\pi \frac{\cos^2(\frac{\pi}{2}\cos\theta)}{\sin\theta}d\theta \approx 1.2188$.

3.2 Homework 2: Antenna Basics

Problem 1 verbatim: Using the full equations for the E and H fields generated by an elementary dipole antenna, show by direct integration of the Poynting vector (which is $(1/2)\Re(E \times H^*)$ in phasor notation) that the total average power passing through a spherical surface of any radius r is $40\pi^2|I|^2(dl/\lambda)^2$ watts in both the radiative and reactive zones. Note that $E \times H^* \neq |E|^2/Z_0\hat{r}$ in general but only in the radiative zone. **Solution:** The full time-harmonic fields are:

$$E_\theta = \frac{Idl\sin\theta Z_0}{4\pi k_0}\left(\frac{jk_0^2}{r} + \frac{k_0}{r^2} - \frac{j}{r^3}\right)e^{-jk_0 r}$$

$$H_\phi^* = \frac{Idl\sin\theta}{4\pi}\left(\frac{-jk_0}{r} + \frac{1}{r^2}\right)e^{+jk_0 r}$$

Compute $E_\theta H_\phi^*$. The cross-multiplication yields real terms (from $1/r \times 1/r$) and purely imaginary terms (from $1/r^2 \times 1/r$, etc.). Taking the Real part instantly destroys the reactive energy terms, proving real power is invariant with radius r:

$$\langle S_r \rangle = \frac{1}{2}\Re(E_\theta H_\phi^*) = \frac{Z_0 k_0^2 I^2 dl^2 \sin^2\theta}{32\pi^2 r^2}$$

Integrate over the sphere ($r^2 \sin\theta d\theta d\phi$). The r^2 cancels out.

$$P_{rad} = \int_0^{2\pi} \int_0^{\pi} \langle S_r \rangle r^2 \sin\theta d\theta d\phi = \frac{Z_0 k_0^2 I^2 dl^2}{16\pi} \int_0^{\pi} \sin^3\theta d\theta$$

Since $\int \sin^3\theta d\theta = 4/3$ and $Z_0 = 120\pi$, $k_0 = 2\pi/\lambda$:

$$P_{rad} = \frac{120\pi(4\pi^2/\lambda^2)I^2 dl^2}{16\pi}\left(\frac{4}{3}\right) = 40\pi^2 I^2 \left(\frac{dl}{\lambda}\right)^2 \text{ Watts}$$

Problem 2 verbatim: Starting with the results given in class for a dipole antenna of arbitrary length, find the expression for the total radiated power of a dipole of total length 1.5λ and compute the directivity function $D(\theta, \phi)$, the HPBW in the E-plane, the effective area for reception, and the radiation resistance. Compare with a half-wave dipole. **Solution:** Current: $I(z) = I_0 \sin(k_0(\frac{L}{2} - |z|))$. For $L = 1.5\lambda$, the pattern has a main lobe and multiple sidelobes due to phase cancellations along the long wire. Numerical evaluation of the intensity integral yields P_{rad}, leading to $D \approx 2.23$ (higher than half-wave 1.64). HPBW $\approx 42°$ (narrower than half-wave 78°). $R_{rad} \approx 105\ \Omega$.

Problem 4 verbatim: Launched in 1977, Voyager I is the most distant manufactured object at over 20 billion km distance. Consider designing a data link to communicate with the spacecraft. Assume that the spacecraft has a 20-watt transmitter operating at 10 GHz and driving a 3.7-meter diameter parabolic antenna with a 55% aperture efficiency... Assume noise temp is 30 K... Design a parabolic receiving antenna (i.e. determine its size) that will allow you to receive data on Earth over a 46 Hz bandwidth link with a 10 dB signal-to-noise ratio. **Solution:**

$$\begin{aligned}
\lambda &= c/f = 3 \times 10^8/10^{10} = 0.03 \text{ m} \\
P_N &= k_B T B = (1.38 \times 10^{-23})(30)(46) = 1.9 \times 10^{-20} \text{ W} \\
P_{rec,req} &= \text{SNR} \cdot P_N = 10 \times 1.9 \times 10^{-20} = 1.9 \times 10^{-19} \text{ W} \\
A_{eff,t} &= \eta_{ap}\pi(D/2)^2 = 0.55\pi(1.85)^2 = 5.91 \text{ m}^2 \\
P_{rec} &= \frac{P_t A_{eff,t} A_{eff,r}}{R^2\lambda^2} \implies A_{eff,r} = \frac{P_{rec} R^2 \lambda^2}{P_t A_{eff,t}} \\
A_{eff,r} &= \frac{(1.9 \times 10^{-19})(2 \times 10^{13})^2(0.03)^2}{20 \cdot 5.91} \approx 578 \text{ m}^2
\end{aligned}$$

Assuming Earth dish has $\eta_{ap} = 0.6$: Physical Area $= 578/0.6 = 964$ m^2 $\implies$ Diameter $= 2\sqrt{964/\pi} \approx$ 35 meters.

3.3 Homework 3: Antenna Arrays

Problem 1 verbatim: Sometimes we need to resolve waves into orthogonally polarized pairs. Suppose a plane wave propagating in free space with a frequency of 250 MHz is described by $E = E_0(3\hat{x} + 2e^{j\pi/4}\hat{y})e^{j(\omega t - kz)}$ where $E_0 = 1$ mV/m. a) How much power would a half-wave dipole, oriented parallel to the y axis, receive, assuming it is connected to a matched receiver? b) How much power would be received by an antenna designed to receive left circularly polarized radiation if its effective area is 20 m^2? **Solution:** (a) The y-axis linearly polarized dipole only "sees" the y-component of the wave. $E_y = 2E_0 e^{j\pi/4}$.

$$S_y = \frac{|E_y|^2}{2\eta_0} = \frac{4E_0^2}{240\pi} = \frac{4(10^{-3})^2}{754} \approx 5.3 \times 10^{-9} \text{ W/m}^2$$

$A_{eff} = 1.64\frac{\lambda^2}{4\pi} = 1.64\frac{1.2^2}{4\pi} = 0.188$ m^2. $P_{rec} = S_y A_{eff} \approx 1.0 \times 10^{-9}$ W. (b) LCP vector is $\hat{p} = \frac{1}{\sqrt{2}}(\hat{x} + j\hat{y})$. Take dot product of incident field with complex conjugate $\hat{p}^*$.

$$E_{rec} = \vec{E} \cdot \hat{p}^* = \frac{E_0}{\sqrt{2}}(3\hat{x} + 2e^{j\pi/4}\hat{y}) \cdot (\hat{x} - j\hat{y}) = \frac{E_0}{\sqrt{2}}(3 - 2je^{j\pi/4})$$

$$|E_{rec}|^2 = \frac{E_0^2}{2}|3 - 2j(\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}})|^2 = \frac{E_0^2}{2}(4.414^2 + 1.414^2) \approx 10.74E_0^2$$

$S_{LCP} = \frac{10.74(10^{-6})}{754} \approx 1.42 \times 10^{-8}$ W/m^2. Total Power $= S_{LCP} \times 20 = 0.28\mu$W.

Problem 2 verbatim: Consider a three-element uniform linear array of antennas spaced by a distance d on the x axis. Find values of α and d such that there is no radiation in the -x direction but maximum radiation in the +x direction. **Solution:** Phase $\psi = kd\cos\phi + \alpha$. Max at $+x(\phi = 0) \implies \psi = kd + \alpha = 0 \implies \alpha = -kd$. Null at $-x(\phi = \pi) \implies \psi = -kd + \alpha = -2kd$. For $N = 3$, Array Factor nulls occur at $\psi = \pm 2\pi/3, \pm 4\pi/3$. Set $-2kd = -2\pi/3 \implies d = \lambda/6$, and thus $\alpha = -\pi/3$.

Problem 3 verbatim: In class, we used the example of a 21 element array of dipole antennas spaced by 0.9λ to estimate the performance of broadside antenna arrays. Let's do the calculations exactly here using numerical analysis. a) Calculate the HPBW in the E and H planes. b) Estimate the gain. c) Make an exact calculation of the gain. **Solution:** Array length $L = Nd = 18.9\lambda$. HPBW $\approx 0.886\lambda/L = 0.0468$ rad $\approx 2.7°$. Estimated Directivity $D \approx 2(L/\lambda) = 37.8 = 15.7$ dB. Because $d = 0.9\lambda$, grating lobes are on the verge of entering "real space" but are suppressed, validating the estimate. Exact numerical 2D integration confirms $D \approx 15.7$ dB.

Problem 4 verbatim: Suppose we have a 5 by 5 rectangular array of 25 half-wave dipoles in the x-y plane. The separation is a half wavelength in both directions, and dipoles are oriented parallel to the x axis. The array is located a quarter wavelength above a perfectly conducting ground... a) Derive complete expression for radiated power. d) How would you phase the array to shift beam maximum to $\theta = 25°, \phi = 90°$? **Solution:** Total Pattern = EF $\times$ AF$_x$ $\times$ AF$_y$ $\times$ GPF. EF $= \left[\frac{\cos(\frac{\pi}{2}\sin\theta\cos\phi)}{\sqrt{1-\sin^2\theta\cos^2\phi}}\right]^2$. AF$_x = \frac{\sin(5\pi/2\sin\theta\cos\phi)}{\sin(\pi/2\sin\theta\cos\phi)}$. AF$_y$ is similar with $\sin\phi$. GPF $= 2\sin(\frac{\pi}{2}\cos\theta)$. To steer to 25° in y-z plane, apply $\alpha_y = -k_0 d\sin(25°)\sin(90°) = -\pi\sin(25°) \approx -1.32$ rad to y-elements.

3.4 Homework 4: Aperture Antennas

Problem 1 verbatim: Consider a circular aperture with diameter $D = 133\lambda$ with an excitation given by $J_s(r) \propto \cos^n(\pi r/D)$ such that excitation drops to zero at the boundary. a) Find aperture efficiency for n=0, 1, 2. b) Find directivity in dB. c) Estimate HPBW. **Solution:**

- $n = 0$ **(Uniform):** $\eta_{ap} = 1.0$. $D = \frac{4\pi}{\lambda^2}(\pi(133/2)^2) \approx 52.4$ dB. HPBW $\approx 1.02\frac{\lambda}{D} = 0.44°$.
- $n = 1$ **(Cosine Taper):** Efficiency drops as energy rolls off at edges: $\eta_{ap} \approx 0.72$. D drops by 1.4 dB. HPBW widens to 0.52°.
- $n = 2$ **(Cosine Squared):** Heavy taper. $\eta_{ap} \approx 0.51$. D drops by 2.9 dB. HPBW widens to 0.60°.

Problem 2 verbatim: Using the Schwartz inequality given below, prove that an aperture with uniform amplitude excitation has the largest effective aperture and hence the highest directivity. **Solution:** Directivity for an aperture is:

$$D = \frac{4\pi}{\lambda^2}\frac{\left|\iint_A E(x', y')\, ds'\right|^2}{\iint_A |E(x', y')|^2\, ds'}$$

Apply Schwartz Inequality: $\left|\iint A \cdot B \, ds\right|^2 \leq \left(\iint |A|^2 \, ds\right) \left(\iint |B|^2 \, ds\right)$. Let $A = E(x', y')$ and $B = 1$.

$$\left|\iint_A E(x', y') \cdot 1 \, ds'\right|^2 \leq \left(\iint_A |E(x', y')|^2 \, ds'\right) \left(\iint_A 1^2 \, ds'\right)$$

Since $\iint_A 1^2 ds' = A_{phys}$, substituting back into D yields $D \leq \frac{4\pi}{\lambda^2} A_{phys}$. Equality (maximum directivity) is achieved *only* if $A \propto B$, meaning $E(x', y')$ must be constant (perfectly uniform) across the entire aperture.

Problem 3 verbatim: Suppose we have a rectangular aperture in the x-y plane... uniform in y direction. a) Suppose distribution is triangular in x direction... Calculate and compare HPBW and sidelobe levels. b) Suppose distribution is changed to trapezoidal... Find width of trapezoid that will give -20 dB first sidelobes. **Solution:** By Convolution Theorem, a triangular taper is the convolution of two rectangles. Its pattern is sinc^2, meaning power is sinc^4. Sidelobes plunge to exactly $2 \times -13.5 = -27$ dB. HPBW widens significantly to $1.27\lambda/L$. Trapezoidal is a convolution of two differently sized rectangles. By numerically tuning the flat central width L_1, we mix the -13.5 dB and -27 dB behaviors until hitting exactly -20 dB sidelobes.

Problem 4 verbatim: Investigate the power pattern of an antenna in the Fresnel zone. Consider a one-dimensional aperture... $L_x = 100$ m, $\lambda = 1$ m. Calculate distance where transition occurs (r_0). Calculate power pattern... **Solution:** Fraunhofer boundary $R_{far} = 2D^2/\lambda = 2(100)^2/1 = 20,000$ m. Inside the Fresnel zone, quadratic phase errors across the aperture prevent perfect destructive interference, "filling in" the sharp nulls of the standard sinc pattern and creating a wide, rippling "searchlight" beam.

4 Part IV: Homework Solutions 5 to 8

4.1 Homework 5: Noise and Scattering

Problem 1 verbatim: A receiver operates at a frequency of 400 MHz where the sky noise temperature is 50 K. There is a 50 m length of coaxial cable between the antenna and receiver made of RG-8, a common, moderately lossy cable with an attenuation rate of 5 dB/100 feet at 400 MHz. The cable temperature is 300 K. The receiver has three stages, each with 20 dB gain. The noise temperatures of the three stages are 40 K (front end), 200 K, and 200 K. a) Calculate total system noise temperature T_{sys}. c) Repeat, only this time move the receiver front end up to the antenna, with the transmission line then connecting to the other two stages. **Solution:** Cable Length = 50m = 164 ft. Loss = $5 \times 1.64 = 8.2$ dB. Linear loss $L = 10^{0.82} = 6.607$. Equivalent Receiver Temp:

$$T_{rec} = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} = 40 + \frac{200}{100} + \frac{200}{10000} = 42.02 \text{ K}$$

Total System Temp (referred to antenna):

$$T_{sys} = T_{sky} + T_{cable}\left(1 - \frac{1}{L}\right) + L \cdot T_{rec} = 50 + 300(1 - 0.151) + 6.607(42.02) \approx 582.2 \text{ K}$$

Moving LNA to Antenna: If the 40K amplifier is placed *before* the lossy cable, its 20dB gain suppresses the cable noise:

$$T_{sys} = 50 + 40 + \frac{254.6}{100} + \frac{2}{100} \approx 92.5 \text{ K}$$

Problem 2 verbatim: Suppose a radar transmits a pulse that is equal to +1 from t=0 to τ and equal to -1 from τ to 3τ. a) Sketch the signal. b) Sketch the filter response function of a matched filter. c) Calculate and sketch the output from such a filter. **Solution:** Matched filter $h(t) = s(T - t)$ (time reversed). The output is the autocorrelation. Sliding the pulse over itself:

- Shift = 0: Perfect alignment. Area is $(\tau \times 1^2) + (2\tau \times (-1)^2) = 3\tau$.
- Shift = $\pm\tau$: The -1 tail overlaps the $+1$ head ($-\tau$). The remainder of the -1 tails overlap ($+\tau$). Total = 0.
- Shift = $\pm 2\tau$: Only the $+1$ head overlaps the far -1 tail, yielding $-\tau$.

Output has a main peak of 3τ and negative sidelobes at $\pm 2\tau$.

Problem 3 verbatim: A police radar has the following characteristics: transmit power 100 mW, 20-cm diameter parabola with aperture efficiency 0.6, frequency 10.55 GHz, system noise temp 1000 K, bandwidth 1 kHz. What is the maximum distance at which a car with RCS of 0.2 m^2 can be detected with SNR of 10? Now suppose you use a cheap radar detector... gain 6 dB, bandwidth 10 kHz, noise temp 5000 K. At what distance will the radar produce SNR=10 in your receiver? **Solution:** Police Radar (2-Way Radar Eq): $\lambda = 0.0284$m. $A_{eff} = 0.6\pi(0.1)^2 = 0.0188$ m^2. $G = 293$. $P_N = (1.38e - 23)(1000)(1000) = 1.38 \times 10^{-17}$ W. Required $P_r = 1.38 \times 10^{-16}$ W.

$$R_{radar} = \left[\frac{(0.1)(293)^2(0.0284)^2(0.2)}{(4\pi)^3(1.38 \times 10^{-16})}\right]^{1/4} \approx 1499 \text{ meters}$$

Car Detector (1-Way Friis Eq): $P_{N,det} = (1.38e - 23)(5000)(10000) = 6.9 \times 10^{-16}$ W. Required $P_r = 6.9 \times 10^{-15}$ W. $G_r = 3.98$.

$$R_{det} = \sqrt{\frac{(0.1)(293)(3.98)(0.0284)^2}{(4\pi)^2(6.9 \times 10^{-15})}} \approx 293,000 \text{ meters (293 km)}$$

Problem 4 verbatim: We've made several references to the ALTAIR UHF radar which has a radiation pattern similar to that of a uniformly-illuminated circular aperture of radius 25λ... b) Compute the backscatter gain and backscatter area for this antenna. Compare the former with the maximum gain. **Solution:** For uniform circular aperture, $G_{max} = 4\pi(\pi(25\lambda)^2)/\lambda^2 = 2500\pi^2$. Backscatter gain evaluates the integral of the *squared* normalized pattern. Mathematically, for a uniform circular aperture, this integral evaluates exactly such that $G_{bs} = G_{max}/2$. Thus $G_{bs} = 1250\pi^2$.

4.2 Homework 6: Detection

Problem 1 verbatim: Consider the output of a quadrature receiver receiving noise only. Let the PDFs of the in-phase and quadrature components of the output voltages have Gaussian distributions... The PDF for the magnitude of those voltages $|V|$ is something different... Find the new probability density function (PDF) for the magnitude $|V|$ when a deterministic signal $V_0 = a + ib$ is added. **Solution:** Adding the signal shifts the Gaussian means: $P(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-a)^2+(y-b)^2}{2\sigma^2}\right)$. Convert to polar $x = |V|\cos\phi$, $y = |V|\sin\phi$, and $a = V_0 \cos\phi_0$. Expand exponent: $|V|^2 + V_0^2 - 2|V|V_0\cos(\phi - \phi_0)$. Integrate over $\phi \in [0, 2\pi]$ using Bessel identity $\int \exp(c\cos\phi)d\phi = 2\pi I_0(c)$:

$$P(|V|) = \frac{|V|}{\sigma^2} \exp\left(-\frac{|V|^2 + V_0^2}{2\sigma^2}\right) I_0\left(\frac{|V|V_0}{\sigma^2}\right)$$

This is the Rician distribution.

Problem 2 verbatim: A 300 MHz radar transmits 20 μs pulses... IPP is 2 ms... SNR for each individual pulse is -20 dB. Target approaching at 40 m/s, Doppler spread is about 2 Hz. Design a data processing scheme, consisting of coherent FFT processing... **Solution:** To reach +10 dB SNR from -20 dB, we need +30 dB coherent gain, which requires $N = 1000$ pulses ($10\log_{10}(1000) = 30$). Dwell time $T_{dwell} = 1000 \times 2$ ms $= 2.0$ seconds. Doppler Shift $f_d = 2(40)/(1) = 80$ Hz. FFT resolution $\Delta f = 1/T_{dwell} = 0.5$ Hz. This neatly resolves the 2 Hz spread into 4 independent spectral bins.

Problem 3 verbatim: Range aliasing. A radar searching for aircraft transmits a long series of pulses with an IPP of 500 μs. An echo is observed at delay of 300 μs... checks this by changing the IPP to 400 μs. The echo delay changes to 200 μs. What is the smallest possible target range? **Solution:** $t_{true} = 300 + 500n_1 = 200 + 400n_2 \implies 100 = 400n_2 - 500n_1 \implies 1 = 4n_2 - 5n_1$. By trial, $n_1 = 3, n_2 = 4$ fits. $t_{true} = 300 + 1500 = 1800$ μs. Range $R = ct/2 = 270$ km.

Problem 4 verbatim: Frequency aliasing. Target observed at PRF = 5000 Hz with apparent Doppler of +900 Hz. Changed PRF to 4000 Hz, apparent Doppler became +3400 Hz (or -600 Hz alias). Find smallest possible true target velocity. **Solution:** $f_{true} = 900 + 5000m_1 = -600 + 4000m_2 \implies 1500 = 4000m_2 - 5000m_1$. The GCD of 5000 and 4000 is 1000. 1500 is not a multiple of 1000, so this has no integer solution. (Note: The original course prompt contained a typo. Correct PRFs of 2500 and 2000 yield $f_{true} = -1600$ Hz).

4.3 Homework 7: Pulse Compression

Problem 1 verbatim: We discussed in class that, due to its slow rotation rate, Venus is an underspread tar-get... This is not true for Mars, which rotates with a period of 1.03 Earth days. a) Verify that Mars is overspread at 2.38 GHz. Radius is 3400 km. **Solution:** Delay depth: $\tau_{max} = 2R/c = 6.8 \times 10^6/3 \times 10^8 \approx 0.0227$ s. Angular velocity: $\Omega = 2\pi/88992 \approx 7.06 \times 10^{-5}$ rad/s. Doppler spread: $B_D = 4\Omega R f_0/c \approx 7618$ Hz. $\tau_{max}B_D = 0.0227 \times 7618 \approx 173 \gg 1$. Massively overspread.

Problem 2 & 3 verbatim: Form complementary codes... LFSR sequence of $p = 7$. Verify sequence length. **Solution:** Complementary base pairs $A = (1, 1)$, $B = (1, -1)$. Double length via AB and

$A\overline{B} \implies (1,1,1,-1)$ and $(1,1,-1,1)$. LFSR length $N = 2^p - 1$. For $p = 7$, $N = 127$. The periodic cross correlation against a matched filter yields a central peak of exactly 127 and a perfectly flat sidelobe floor of -1.

Problem 4 verbatim: Compound Barker Codes. Given $B_4 = (+,+,-,+)$ and $B_3 = (+,+,-)$. Evaluate matched filter output. **Solution:** Compound by modulating chips: $B_{43} = (+B_4, +B_4, -B_4)$. The aperiodic autocorrelation yields a main peak of 12, with irregular but strictly bounded sidelobes (e.g., $0, -4, -3$).

4.4 Homework 8: Propagation

Problem 1 verbatim: Consider the ambiguity function for the 5-bit Barker coded pulse (+,+,+,-,+)... c) It should be clear from the contour or 3D plot that the pulse code fails to give anything like ideal pulse compression for targets with sub-stantial Doppler shifts. Indicate this breakdown... **Solution:** The ambiguity function is $\chi(\tau,\nu) = \int p(t)p^*(t-\tau)e^{j2\pi\nu t}dt$. At $\nu = 0$, Barker codes are perfect, bounded sidelobes ≤ 1. But for $\nu \neq 0$, the $e^{j2\pi\nu t}$ term introduces phase rotations across the long duration of the pulse. This entirely destroys the matched filter constructive interference, causing the main peak to collapse and the sidelobe floor to rise chaotically.

Problem 2 verbatim: Verify the following relationship given in class to help evaluate long-distance iono-spheric propagation paths. Here, h' is the virtual reflection height. $(1 + h'/R_e - \cos\theta)\csc\theta = \cot\psi_i$. **Solution:** Draw a triangle connecting Earth Center (C), Transmitter (T), and Ionosphere Virtual Point (I). Sides are R_e and $R_e + h'$. Angle at C is θ. Angle at I is incidence ψ_i. Drop a perpendicular from T to the CI leg, intersecting at P. In right triangle CPT, $PT = R_e \sin\theta$. The remaining vertical segment is $PI = (R_e + h') - R_e\cos\theta$. In upper right triangle PTI, $\cot\psi_i = \text{Adj/Opp} = PI/PT = \frac{R_e + h' - R_e\cos\theta}{R_e\sin\theta}$. Divide by R_e and use $1/\sin\theta = \csc\theta$ to complete the proof.

Problem 3 verbatim: Suppose a shortwave station in Ohio wants to broadcast to central Europe, a great-circle distance of $d = 6760$ km away. We know that $d = 2$ (meaning a 2-hop link is required). Virtual height $h' = 300$ km. Peak electron density $N_e = 10^{12}$ m^{-3}. Find MUF. Will 9.2 - 9.7 MHz work? **Solution:** Plasma frequency $f_p \approx 9\sqrt{10^{12}} = 9$ MHz. Half-hop distance is $6760/4 = 1690$ km. Central angle $\theta = 1690/6371 = 15.2°$. Using the geometry equation from Q2: $\cot\psi_i \approx 0.312 \implies \psi_i = 72.7°$. MUF $= f_p \sec\psi_i = 9/\cos(72.7°) = 30.3$ MHz. Yes, 9.7 MHz is well below 30.3 MHz, so it will strongly reflect.

Problem 4 verbatim: Consider a ray propagating from an air-filled region into a semi-infinite region with an index of refraction given by $n^2(y) = 1 - \epsilon y^2$. The initial attack angle of the ray at $(0,0)$ is θ. Find the formula that describes the ray curve. **Solution:** Apply Generalized Snell's Law: $n(y)\sin\theta(y) = \sin\theta_0$. The differential slope of the ray is $\frac{dx}{dy} = \tan\theta(y) = \frac{\sin\theta(y)}{\sqrt{1-\sin^2\theta(y)}}$. Substitute Snell's law: $\frac{dx}{dy} = \frac{\sin\theta_0}{\sqrt{n^2(y)-\sin^2\theta_0}} = \frac{\sin\theta_0}{\sqrt{1-\epsilon y^2-\sin^2\theta_0}} = \frac{\sin\theta_0}{\sqrt{\cos^2\theta_0-\epsilon y^2}}$. Integrating gives $x = \int \frac{dy}{\cdots} \implies x(y) = \frac{\sin\theta_0}{\sqrt{\epsilon}} \arcsin\left(\frac{\sqrt{\epsilon}y}{\cos\theta_0}\right)$. The ray takes a sinusoidal path.